

Problem J Big Two

Time limit: 3 seconds

Memory limit: 1024 megabytes

Problem Description

Experts say that playing cards can help slow down memory decline, so the elderly residents in the nursing home often gather to play Big Two. However, since they take a long time to read their cards, each game drags on for too long, and some players even fall asleep in the middle of it, much to the caregivers' frustration.

To reduce these unexpected nap times, the director has asked you to design a program that helps the residents quickly determine how many possible straights they have in their hands.

The rules for straights in Big Two are as follows:

- A straight consists of five consecutive cards.
- Valid straights include: 10 J Q K A.

Input Format

Your program is to write to standard output. The input may contain multiple test cases. Each test case consists of two lines: the first line contains an integer P , which indicates the number of cards in the player's hand; the second line contains $C_1, C_2, \dots, C_P$, separated by spaces, representing the cards in the player's hand. The program ends when P equals '0'.

Output Format

Your program is to write to standard output. If the player's hand can form a valid straight, output the five cards of the straight separated by spaces, with each straight printed on a separate line. If there are multiple straights, output them in ascending order based on the value of the first card, with the sequence ordered as A, 2, 3, ..., J, Q, K. If no straight can be formed, output 'NONE'. When the number of cards in the input is '0', the program terminates. Please see the sample output.

Technical Specification

- $1 \leq P \leq 13$
- $C_i \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, A, J, Q, K\}$

Sample Input 1

```
5
9 8 5 3 K
6
A 2 3 4 5 6
```

Sample Output 1

```
NONE
A 2 3 4 5
2 3 4 5 6
```

0